

Structured summary: Informative dropout and trial-design choice in aggregated N-of-1 biomarker-validation trials

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August 15, 2026

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This document is a structured, section-by-section summary of the key points argued in the companion manuscript, “Informative dropout and trial-design choice in aggregated N-of-1 biomarker-validation trials.” It is provided as a supplementary reading aid and does not stand on its own; all notation, derivations, simulation detail, and citations are in the main manuscript.

0.1 A claim about how N-of-1 trials should account for dropout

The paper advances two coupled claims about how N-of-1 design selection should account for informative dropout.

- The four design families (OL, OL+BDC, traditional crossover, hybrid) differ in dropout robustness in ways the published literature has not adequately characterized.
- Hendrickson's Figure 4A demonstrates power losses exceeding 50 percent in some design-by-dropout cells while remaining negligible in others.
- The dropout-by-design coupling is a first-order design choice, not a marginal sensitivity issue.

The first claim is that design families differ in dropout robustness in ways the published literature has not adequately characterized.

- Power loss from a realistic dropout pattern can exceed 50 percent in some design-by-dropout cells.
- Design rankings reverse depending on the dropout pattern.
- This is a first-order design choice, not a marginal sensitivity issue.

The second claim is that the dropout mechanism, not merely the rate, is the methodologically relevant object for design selection.

- Balanced, worsening-driven, and improvement-driven mechanisms interact with design families in qualitatively different ways.
- Trial designers should evaluate designs against the anticipated dropout mechanism rather than a generic rate.
- The mechanism is the primary object; the rate is relatively superficial.

The paper imports the mature missing-data taxonomy into the N-of-1 design-comparative context for the first time.

- The missing completely at random (MCAR), missing at random (MAR), and missing not at random (MNAR) taxonomy and the mixed-effects model for repeated measures (MMRM) have been worked through for parallel-group RCTs but not for N-of-1 designs comparatively.
- ICH E9 (R1) formalizes estimand-aware handling of intercurrent events including dropout.
- The present paper undertakes the import of this machinery into the N-of-1 design-comparative setting.

0.2 Why the question matters

Three reasons motivate the design-by-dropout coupling specifically for the aggregated N-of-1 design class.

- Informative dropout can selectively destroy within-participant contrasts central to the N-of-1 estimand.

- Patient self-selection is correlated with latent biomarker-response components that are partly signal, not nuisance.
- The four canonical designs differ in their phase-specific vulnerability windows.

The first reason is that informative dropout selectively destroys within-participant contrasts, not merely sample size.

- Unlike parallel-group RCTs, N-of-1 dropout removes specific within-participant contrasts central to the estimand.
- A participant lost before the blinded discontinuation phase contributes nothing to the BR-versus-PB separation.
- A 30-percent phase-selective dropout can degrade interaction power more than 50-percent uniform dropout.

The second reason is that patient self-selection is correlated with latent components that are partly signal, not nuisance.

- High-BR patients tend to remain on drug; high-PB patients may drop at discontinuation; negative-TV patients may drop from worsening.
- Each mechanism produces a different dropout pattern that interacts differently with the design family.
- The worsening- and improvement-driven patterns operationalize the responder-versus-non-responder distinction the biomarker literature is validating.

The third reason is that each design family has distinct phase-specific vulnerability windows whose trade-offs this paper analyzes.

- OL is most dropout-robust but cannot identify the BR-versus-PB split required for the predictive-biomarker question.
- OL+BDC introduces a discontinuation-phase window where responders may drop from symptom return.
- The hybrid design has the largest vulnerability surface; the trade-off between identifiability and dropout vulnerability is the central design choice.

0.3 What the published N-of-1 literature does and does not do

The methodological N-of-1 corpus addresses dropout generically but does not engage the design-by-dropout coupling this paper centers on.

- Lillie et al. and Vohra et al. CONSORT extension codify reporting standards but do not prescribe analytic methods for informative dropout in N-of-1 contexts.
- Systematic reviews document that dropout rates are reported without specifying mechanism or analytic treatment.
- Dropout is one of the leading methodological weaknesses in N-of-1 reporting audits.

Existing N-of-1 reporting standards codify dropout disclosure but do not prescribe analytic methods for informative dropout.

- Lillie et al. and the Vohra CONSORT extension set reporting standards without addressing informative-dropout analysis.
- Systematic reviews find that substantial proportions of N-of-1 studies report rates without specifying mechanism or analytic treatment.
- Dropout is one of the leading methodological weaknesses in N-of-1 reporting audits.

Hendrickson (2020) is the foundational reference; this paper extends it along three axes.

- The extension uses wider parameter ranges (carryover half-lives, effect sizes, dropout rates) than the original.
- It examines sensitivity to dropout-pattern misspecification at the analysis stage, which the original assumed away.
- It offers an explicit decomposition of the happy-accident randomization-path selection effect, scoped for path-stratified production testing.

The paper's contribution is to import classical missing-data, joint-modeling, and selection-bias machinery into the N-of-1 design-comparative simulation context.

- Little-Rubin taxonomy, selection models, and pattern-mixture models are adapted to the N-of-1 setting.
- Hazard-coupled-to-longitudinal-trajectory joint-modeling underwrites the Hendrickson dropout model and its extension.
- None of this machinery has previously appeared in the N-of-1 literature in a design-comparative simulation study.

0.4 What this paper contributes

The paper makes four contributions across Sections 2 through 6.

- Section 2 formalizes the informative-dropout model and places it in the missing-data taxonomy.
- Section 3 characterizes each design family's phase-specific vulnerability windows.
- Sections 4 and 5 conduct and report a Monte Carlo simulation crossing four designs against five dropout patterns.

Section 2 formalizes the hazard-based informative-dropout model and establishes its MNAR character explicitly.

- The Hendrickson hazard model is placed in the MCAR/MAR/MNAR taxonomy and shown to be MNAR.
- Four canonical dropout patterns plus a no-dropout reference are defined as cells in the (β_0, β_1) hazard-parameter space.
- The operational definitions fix the simulation's data-generating mechanism for the subsequent studies.

137 **Section 3 identifies the phase-specific dropout-vulnerability windows and at-risk**
138 **within-participant contrasts for each design family.**

- 139 • Each of the four design families is characterized by its exposure to each dropout pattern
140 per phase.
- 141 • Within-participant contrasts that are at risk of removal under each pattern are identified
142 explicitly.
- 143 • The characterization motivates the design-by-dropout-pattern structure of the simula-
144 tion in Sections 4 and 5.

145 **Sections 4 and 5 report a Monte Carlo study crossing four designs against five**
146 **dropout patterns at three sample sizes and three effect sizes.**

- 147 • Performance measures include power, Type I error, bias, empirical SE, and path-
148 conditional power decomposition.
- 149 • Reporting follows the Morris (2019) ADEMP framework with Monte Carlo standard
150 errors (MCSE) alongside every estimate.
- 151 • The randomization-path decomposition quantifies the happy-accident selection effect.

152 **Section 6 synthesises a recommendation table mapping dropout mechanism class**
153 **to preferred N-of-1 design choice.**

- 154 • Three mechanism classes (balanced, worsening-driven, improvement-driven) are mapped
155 to preferred designs indexed by anticipated dropout rate.
- 156 • The table is presented as practical guidance for trial designers selecting among the four
157 canonical N-of-1 families.
- 158 • Section 7 discusses implications for trial-design selection and pmsimstats operating char-
159 acteristics.

160 **Section 7 discusses implications for trial-design selection and pmsimstats operat-**
161 **ing characteristics.**

- 162 • The discussion addresses design selection for biomarker-validation N-of-1 trials.
- 163 • Operating characteristics of the pmsimstats simulation program are reviewed.

164 **1 Informative-dropout model**

165 **The censoring hazard combines a time-only ignorable component with a symptom-**
166 **trajectory-biased component, made absorbing by a monotone carry-forward pass.**

- 167 • The per-timepoint censoring probability is the union of two independent draws: a flat
168 component scaling with elapsed time and a biased component scaling with the convex-
169 transformed symptom change since baseline.
- 170 • Two hazard parameters govern the split: β_0 sets the overall dropout fraction and β_1 the
171 proportion of that fraction routed through the symptom-biased component, with the
172 convexity exponent e_{b1} fixed at two.

- A second carry-forward pass renders dropout absorbing: once a participant is censored at a timepoint, all later timepoints for that participant are censored.

Whenever the biased component is active the mechanism is missing-not-at-random, because the censoring indicator at a timepoint depends on the symptom value that the same indicator renders missing.

- The flat component depends only on elapsed time, a fully observed design variable, and is therefore ignorable with respect to the latent response.
- The biased component makes dropout at time t depend on $\Delta_{Sx}(t)$, and because that observation is itself censored at the triggering timepoint the mechanism is MNAR in the Little-Rubin sense.
- The departure from ignorability is controlled by β_1 : at $\beta_1 = 0$ the mechanism reduces to the ignorable limit, and the MNAR departure increases monotonically with β_1 .

Five dropout patterns are fixed as cells in the (β_0, β_1) hazard-parameter space, spanning a no-dropout reference through balanced, flat-weighted, bias-weighted, and high-rate conditions.

- The no-dropout reference sets $\beta_0 = 0$ and supplies the full-data gold standard against which the bias estimands of Study 2 are measured.
- Balanced ($\beta_0 = 0.05, \beta_1 = 0.5$), more-of-flat ($\beta_1 = 0.2$), and more-of-biased ($\beta_1 = 0.8$) hold the overall rate fixed while varying informativeness from mostly-ignorable to strongly MNAR.
- High-dropout ($\beta_0 = 0.15, \beta_1 = 0.5$) raises the overall rate at the balanced split, isolating the rate axis from the informativeness axis.

2 The four design families and their vulnerability windows

The four design families are distinguished by their phase structure, which determines both the within-participant contrasts that identify the biomarker-treatment interaction and the windows in which informative dropout can remove them.

- Each design is a set of randomization paths through an ordered sequence of on-drug and off-drug phases, encoded by the `ondrug` path vectors of `buildtrialdesign`.
- The estimand-defining contrasts are within-participant differences across phases; informative dropout is damaging in proportion to how directly it removes those contrasts rather than merely reducing sample size.
- The four designs are ordered by an increasing number of identifying contrasts at risk, and therefore by an increasing number of vulnerability windows, from open-label through the hybrid N-of-1 design.

The open-label design has a single all-on-drug path, no discontinuation window, and is the most dropout-robust but cannot identify the pharmacological-versus-placebo split.

- The single path carries only an on-drug-versus-baseline contrast, which is confounded with natural history and cannot separate the *BR* and *PB* components.
- With no off-drug phase there is no discontinuation-driven vulnerability window; dropout reduces only the precision of the on-drug slope.
- The design is the robustness benchmark but is the wrong design for the predictive-biomarker question.

The open-label-plus-blinded-discontinuation design adds a silent-placebo phase that identifies the pharmacological-versus-placebo split but opens a discontinuation window in which worsening-driven dropout removes exactly that contrast.

- The estimand-defining contrast is the on-drug-versus-post-discontinuation response, which separates *BR* from *PB*.
- Treatment-responders whose symptoms return on the silently switched placebo may drop, mistaking the return for trial failure, which is a worsening-driven dropout located in the discontinuation phase.
- Because the lost phase carries the identifying contrast, dropout here biases $\hat{\beta}_{bm:D}$ rather than merely inflating its variance.

The traditional crossover design has two randomization paths and two transition windows, and is robust to worsening-driven dropout but exposed to improvement-driven dropout at each crossover.

- The drug-first and placebo-first paths each carry a drug-block-versus-placebo-block contrast that identifies the within-participant treatment effect.
- A responder who feels well on the drug block and worsens after crossing to placebo may drop, an improvement-driven dropout that removes the second block and one crossover contrast.
- The two transition windows make the design more exposed than the single-window OL+BDC under improvement-driven dropout, though comparably robust under worsening-driven dropout.

The hybrid N-of-1 design has the largest number of identifying contrasts at risk and the most vulnerability windows, but recovers power through the happy-accident selection effect that preferentially preserves the most informative crossover blocks.

- An open-label run-in followed by a randomized multi-block crossover gives four randomization paths and the widest set of vulnerability windows.
- Participants with a clear open-label response are resolved early, so those continuing to the later randomized blocks are enriched for inconclusive open-label response.
- Biased dropout therefore preferentially preserves the most informative crossover blocks, partially offsetting the raw power loss from the larger vulnerability surface.

3 Simulation design

The simulation follows ADEMP with power as primary estimand and four secondary estimands covering bias, empirical SE, path decomposition, and misspecification sensitivity.

- Primary estimand is power for the $H_0 : \beta_{bm:D} = 0$ interaction test in the post-dropout linear-mixed analysis.
- Bias is reported in two modes: relative to the calibrated true value and relative to the full-data gold standard.
- Secondary estimands include path-conditional power (Study 3) and dropout-pattern misspecification sensitivity (Study 4).

Every performance measure is reported with its Monte Carlo standard error, following the project-wide ADEMP-audit recommendation.

- Power and Type I error MCSE follow $\sqrt{\hat{\pi}(1 - \hat{\pi})/n_{\text{reps}}}$.
- Bias and empirical SE of $\hat{\beta}_{bm:D}$ carry the standard MCSE of the within-replicate mean.
- Reporting MCSEs alongside every estimate is a project-wide ADEMP-audit requirement.

Four pre-registered studies fix the simulation scope before production runs: power, bias, path decomposition, and misspecification sensitivity.

- Study 1 (180 cells) reproduces and extends Hendrickson Figure 4A for power as a function of design and dropout pattern.
- Study 2 (120 cells) addresses two-mode bias quantification; Study 3 (12 cells) examines the happy-accident path decomposition.
- Study 4 (60 cells) evaluates sensitivity to dropout-pattern misspecification at the analysis stage.

4 Results

The prototype ran 500 replicates per cell across 16 cells with 100 percent convergence and achieved dropout fractions close to targets.

- 8-core `parallel::mclapply` completed 8,000 fits with 100 percent convergence.
- Cell MCSE on power runs 0.008–0.019, widest in the binding 0.85 regime; at matched N no cell is saturated.
- Achieved dropout fractions tracked targets: approximately 0.25 for `MCAR_25` and `biased_25`, approximately 0.40 for `biased_40`.

4.1 Power across designs and dropout patterns

At a matched total of $N = 70$ the Hendrickson happy-accident ordering holds, but the hybrid margin over OL+BDC is small.

- No dropout: hybrid 0.968 > OL+BDC 0.950 > traditional crossover 0.854 > OL 0.088.
- The ordering is the same at every dropout pattern, so it is a property of the designs and not of one dropout condition.
- The hybrid-over-OL+BDC gap reaches significance only under 25 percent uninformative dropout (5.4 points, $z = 3.59$); at no dropout (1.8, $z = 1.44$) and at 40 percent biased dropout (1.6, $z = 0.85$) the two are not separated.

The OL row carries almost no interaction information, which is a structural property of the design under the mean-moderation architecture rather than a dropout effect.

- Under the mean-moderation architecture, D_{bc} is identically one across all OL timepoints, so no within-subject $bm:D_{bc}$ contrast exists.
- Rejection rates sit at 0.086–0.112 across the four dropout cells, an order of magnitude below the other designs and flat with respect to dropout.
- They are nonetheless about twice nominal alpha, so the row is not a clean null: with the interaction term aliased, the reported rate is not a valid test size and should be read as an information floor, not as either power or Type I error.

At matched N the hybrid and OL+BDC designs are equally vulnerable to informative dropout; only the crossover design is clearly worse.

- Absolute power loss from no-dropout to `biased_40`: hybrid 5.8 pp, OL+BDC 5.6 pp, traditional crossover 8.6 pp.
- Hybrid and OL+BDC are indistinguishable on this measure; the crossover loss is about half again as large.
- The hybrid design's apparent near-immunity in the earlier unmatched run (0.4 pp) was an artifact of its power sitting against the ceiling at twice the sample, not a structural property.

4.2 Bias of the biomarker-treatment interaction estimate

The mean estimated interaction coefficient is stable across dropout cells within each design, indicating no systematic bias in the estimand.

- $\hat{\beta}_{bm:D_{bc}}$ shows no systematic trend across none, MCAR_25, biased_25, and biased_40 cells within any design.
- Stability holds for the three powered designs (OL+BDC, crossover, hybrid); the OL design is uninformative on bias.
- The absence of bias in the estimand implies the power degradation traces to standard-error inflation, not bias.

Power degradation under informative dropout traces to standard-error inflation, not bias in the estimand: empirical SEs expand modestly under biased_40 dropout.

- The largest within-design bias shift (approximately 0.04 for traditional crossover) is within cell-level MCSE.
- Empirical SEs expand under `biased_40`: OL+BDC 0.628 to 0.660, crossover 0.766 to 0.779, hybrid 0.411 to 0.454.
- The power loss in Section 5.1 is therefore attributable to SE inflation rather than systematic bias.

4.3 Randomization-path decomposition

Path-stratified decomposition of the happy-accident effect is scoped for a production rerun; the current prototype provides only marginal results consistent with the hypothesis.

- The current prototype aggregates across paths within each design and cannot attribute robustness to specific surviving paths.
- The Hendrickson happy-accident hypothesis is consistent with the marginal results but requires path-tracked data for a formal test.
- The follow-up extension requires recording `path_id` per replicate and adding path-stratified $\hat{\beta}_{bm:D_{bc}}$ summary logic.

4.4 Sensitivity to dropout-pattern misspecification

The `biased_25` versus `MCAR_25` contrast isolates the bias mechanism from the rate effect; within-design differences are small and statistically indistinguishable at 500 replicates per cell.

- OL+BDC: `biased_25` minus `MCAR_25` equals +1.2 pp (CI [-2.4, +4.8]); crossover: +1.4 pp (CI [-3.5, +6.3]).
- Hybrid: -1.8 pp; OL: -0.8 pp on aliased cells; all differences lie within or near 1.96 cell-MCSE.
- Resolving the distinguishability question would require approximately 2,000 replicates per cell at the current MCSE.

The dominant power determinant is dropout rate rather than bias mechanism; mechanism effects are not statistically distinguishable at 500 replicates per cell.

- Moving from 25% to 40% biased dropout costs 3.0 pp (OL+BDC), 5.6 pp (crossover), 3.8 pp (hybrid).
- The crossover rate-effect loss is clearly detectable above MCSE; the OL+BDC and hybrid losses are borderline.
- Resolving the `biased_25` versus `MCAR_25` mechanism question requires approximately 2,000 replicates per cell, deferred to a production rerun.

5 Design-by-dropout recommendation

Three classes of anticipated dropout mechanism are mapped to preferred N-of-1 design choices based on the Section 5 simulation results.

- The mapping is indexed by anticipated overall dropout rate within each mechanism class.
- The recommendation draws directly on the power and bias results from the 16-cell prototype simulation.
- The table is intended as practical guidance for trial designers selecting among the four canonical N-of-1 families.

The dominant recommendation message is that the hybrid and OL+BDC designs are interchangeable on these criteria and the traditional crossover is the one to avoid under heavy dropout.

- Hybrid and OL+BDC are not separable on power at matched N except under 25 percent uninformative dropout, and are indistinguishable on dropout loss, so the choice between them mostly turns on cost and complexity.
- Traditional crossover loses approximately 9 percentage points under `biased_40` and is the least dropout-robust powered design.
- The OL design lacks the within-subject D_{bc} contrast under the mean-moderation architecture and does not warrant inclusion in the primary recommendation.

5.1 Prototype Monte Carlo confirmation

The 500-replicate prototype confirms 100 percent convergence across all 8,000 fits with achieved dropout fractions close to targets.

- Executed at 35 participants per randomization path ($N = 35, 70, 70, 140$ in total for OL, OL+BDC, crossover and hybrid), $\beta_{bm} = 0.45$, the mean-moderation architecture on 8 cores in 421 s with 100 percent convergence.
- Achieved dropout: approximately 0.25 for `MCAR_25` and `biased_25`; approximately 0.40 for `biased_40`.
- Cell MCSE on power runs 0.008–0.019; at matched N no cell is saturated.

The prototype reproduces the Hendrickson happy-accident power ordering at matched N : `hybrid` > `OL+BDC` > `crossover` > `OL`.

- Power ordering without dropout: hybrid 0.968, OL+BDC 0.950, crossover 0.854, OL 0.088.
- The branching after week 8 gives the hybrid design within-subject and across-path D_{bc} contrasts, placing it in the top pair on equal sample.
- The ordering holds across all four dropout-pattern cells, though the hybrid-versus-OL+BDC gap is resolvable only under 25 percent uninformative dropout.

The OL design's nominal-alpha rejection rate is a structural property of the mean-

391 **moderation architecture, not an artifact of dropout, and serves as an implicit null**
392 **check.**

- 393 • Under the mean-moderation architecture, $D_{bc} \equiv 1$ for all OL timepoints, eliminating
394 any within-subject $bm:D_{bc}$ contrast.
- 395 • The fitter has no interaction information, so rejection rates collapse to nominal alpha
396 across all dropout conditions.
- 397 • This confirms the OL row is a structural null check rather than evidence of dropout
398 robustness.

399 **Dropout resilience is dominated by the rate effect, not the bias mechanism; the**
400 **linear-mixed model with corCAR1 is relatively unbiased under informative dropout**
401 **in this regime.**

- 402 • Absolute power loss from no-dropout to `biased_40`: hybrid 0.004, OL+BDC 0.056,
403 crossover 0.086.
- 404 • The `biased_25` vs. `MCAR_25` mechanism contrast is within or barely outside 1.96 MCSE;
405 the rate increase from 25 to 40 percent is the dominant effect.
- 406 • $\hat{\beta}_{bm:D_{bc}}$ is stable (-2.45 to -2.48), confirming power degradation traces to SE inflation
407 rather than bias in the estimand.

408 **Two ADEMP caveats limit the prototype's scope: no formal null cells for non-OL**
409 **designs, and single-architecture coverage.**

- 410 • No $\beta_{bm} = 0$ cells were run for OL+BDC, crossover, or hybrid; production reruns should
411 add explicit null cells for each.
- 412 • The prototype is single-architecture (the mean-moderation architecture); under the
413 covariance-moderation architecture the OL row carries information via the covariance
414 structure.
- 415 • Resolving the `biased_25` vs. `MCAR_25` distinguishability question requires approximately
416 2,000 replicates per cell.

417 6 Conclusions

418 **The conclusions distil three findings: design-by-dropout differentiation, rate dom-**
419 **inance over mechanism, and SE inflation as the power-loss mechanism.**

- 420 • The four N-of-1 design families differ materially in dropout robustness in ways the
421 published literature has not characterized.
- 422 • Dropout rate dominates mechanism as the determinant of power loss; mechanism be-
423 comes load-bearing only when rates are comparable.
- 424 • Estimand stability under informative dropout is reassuring but also a warning: coverage
425 degrades as SEs widen without the model acknowledging it.

426 **First conclusion: the traditional crossover is the least dropout-robust of the**
427 **powered designs, while the hybrid and OL+BDC designs are equally robust at**

428 **matched sample size.**

- 429 • At a matched total of $N = 70$, $\beta_{bm} = 0.45$, the mean-moderation architecture: hybrid
430 loses 5.8 pp, OL+BDC 5.6 pp and crossover 8.6 pp under **biased_40**.
- 431 • OL lacks the within-subject D_{bc} contrast under the mean-moderation architecture and
432 is not viable for the biomarker-interaction question.
- 433 • The hybrid design's apparent near-immunity in an earlier unmatched run was a ceiling
434 effect at twice the sample, so the happy-accident argument survives as a power claim
435 but not as a robustness claim.

436 **Second conclusion: dropout rate is the dominant power determinant; mechanism**
437 **is load-bearing only when rates are comparable across competing design candi-**
438 **dates.**

- 439 • The **biased_25** vs. **MCAR_25** contrast at fixed 25 percent dropout is below MCSE detec-
440 tion at 500 replicates.
- 441 • Moving to 40 percent biased dropout produces losses in all three powered designs, clear-
442 est in the crossover.
- 443 • Trial designers should evaluate designs against anticipated dropout rate first, and mech-
444 anism only secondarily.

445 **Third conclusion: estimand stability is reassuring but also a coverage warning,**
446 **since the $\hat{\beta}_{bm:D_{bc}}$ distribution widens without the model acknowledging it.**

- 447 • The largest within-design shift in $\hat{\beta}_{bm:D_{bc}}$ is below 0.04 across the no-dropout to
448 **biased_40** range.
- 449 • Power degradation traces to SE inflation; the inferential target is preserved under infor-
450 mative dropout in this regime.
- 451 • Nominal CI coverage may understate true uncertainty: the coefficient distribution
452 widens but the model does not acknowledge it.

453 **Three production-grade extensions are scoped: path tracking, the covariance-**
454 **moderation architecture coverage, and higher replicate counts to resolve the**
455 **biased-versus-MCAR distinguishability question.**

- 456 • Per-path-id tracking is needed for the randomization-path decomposition of the happy-
457 accident effect.
- 458 • The covariance-moderation architecture coverage is needed to interpret the OL row as
459 a power estimate rather than a structural null check.
- 460 • Resolving the **biased_25** vs. **MCAR_25** question requires 2,000 or more replicates per cell.